

Question 1. Log Mining and Analysis [10 marks, set by Shuo Zhou]

Please carefully read the [assignment brief](#) before starting to complete the assignment.

Prerequisites:

Before attempting this question, you are expected to have completed Lab 1 and Lab 2.

Data:

Download the NASA access log for July 1995 using:

```
Shell
wget ftp://ita.ee.lbl.gov/traces/NASA_access_log_Jul95.gz
```

Unzip the file, for example:

```
Shell
gunzip NASA_access_log_Jul95.gz
```

Then move the file to the “Data” folder, for example,
'/users/your_username/com6012/ScalableML/Data'.

Definitions

For this assignment:

- A request = one log entry.
- A unique host = a unique hostname appearing in the host field.
- A UK academic host = hostname ending with .ac.uk
- A UK company host = hostname ending with .co.uk
- A UK government host = hostname ending with .gov.uk
- For institution/company level: Use the last three labels of the hostname (e.g., shef.ac.uk, cam.ac.uk, bt.co.uk, nhs.gov.uk).

Tasks:

A. For each of the following categories:

- .ac.uk (UK academic institutions)
- .co.uk (UK companies)
- .gov.uk (UK government departments)

Compute:

- Total number of requests
- Number of unique hostnames

Present your results as a table in your report with columns:

| Sector | Total Requests | Unique Hosts |

[1 mark]

B. Using UK academic hosts (*.ac.uk):

1. Compute the total number of requests from the University of Sheffield domain (ending with shef.ac.uk).
2. Find out how many other UK academic institutions made more requests than Sheffield.
3. Create a DataFrame to store the above institutions with more requests than Sheffield, along with the number of requests for each. Then save the DataFrame as a CSV file.

You are expected to explain:

- How do you extract an “institution domain”
- How do you aggregate requests at the institution level

To demonstrate this, you may include your code segments and the resulting table in your report.

[2 marks]

C. Using UK company hosts (*.co.uk):

- Identify the 9 most active UK companies by total number of requests.
- Produce a visualisation showing the number of requests for each of the top 9 companies compared with all the other companies (10 items in total).

You are free to choose an appropriate visualisation type. Your figure must:

- be clearly labelled
- be interpretable with minimum explanation (caption)
- be created by your Python program and saved as a .png/.jpg file

Briefly justify your design choice and include the figure in your report.

[2 marks]

D. Create heatmaps showing access patterns for:

1. The most active UK academic institution domain (*.ac.uk)
2. The most active UK company domain (*.co.uk)
3. All UK government domains combined (gov.uk)

(“Most active” = highest total number of requests.)

For each heatmap:

- x-axis: Day of month (1-28)
- y-axis: Hour of day (0–23)
- Colour intensity: Number of requests

Produce three separate heatmaps using your Python program and save them as .png/.jpg files. These figures should be included in your report.

[3 marks]

E. Select two observations from Tasks A–D that you find interesting.

For each observation, write at least three sentences addressing the following questions:

1. What is the observation?
2. What are the possible causes?
3. How could this insight be useful to NASA?

Each observation must refer to specific results or figures from your analysis.

[2 marks]